Scale-Factor Calibration and Earth-Rotation Sensing with a Passive Ring-Laser Gyroscope

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We characterize a passive ring-laser gyroscope and evaluate its scale factor alongside its sensitivity to the Earth's rotation. We determine the cavity's free spectral range to be 300(10) MHz and its perimeter to be 100(3) cm. Through cavity ringdown measurements, we calculate a mean finesse of $74(26) \times 10^3$. We calibrate the gyroscope's scale factor using a motor-driven rotation table, extracting a value of $237\,888(11278)\,\mathrm{Hz\,rad^{-1}\,s^{-1}}$ via a signed free-intercept linear fit, which aligns with theoretical expectations. An Allan-deviation analysis identifies a white-noise-limited angle random walk of $0.144\,\mathrm{mrad}/\sqrt{\mathrm{s}}$ in the most stable measurement window. Finally, we attempt to resolve the Earth's rotation rate but demonstrate that direction-independent systematic offsets and a dominant lock-in effect entirely obscure the terrestrial signal.

I. INTRODUCTION

The Sagnac effect forms the foundation of modern optical gyroscopes, enabling highly precise rotation measurements by evaluating the phase shift between counter-propagating light beams. In this advanced laboratory course, we investigate a passive free-space ring-laser gyroscope [1]. Unlike active ring lasers, passive gyroscopes couple external laser light into an optical resonator, isolating the gain medium from the rotation-sensing cavity. We employ the Pound-Drever-Hall locking technique to continuously stabilize the counter-propagating laser beams to the cavity resonances. By overlapping the transmitted beams on a photodiode, we extract the Sagnac beat frequency, which acts as a direct observable for the applied rotation rate.

This report structures our experimental characterization of the gyroscope. We begin by analyzing the fundamental properties of the optical cavity, identifying its free spectral range, transverse mode profile, and overall finesse. Next, we calibrate the gyroscope's scale factor by applying controlled angular velocities via a motorized rotation stage. We subsequently evaluate the sensor's underlying stability and noise characteristics using the Allan deviation. In the final phase, we test the setup's capacity to measure the Earth's true rotation rate and analyze the severe restrictions imposed by the lock-in effect.

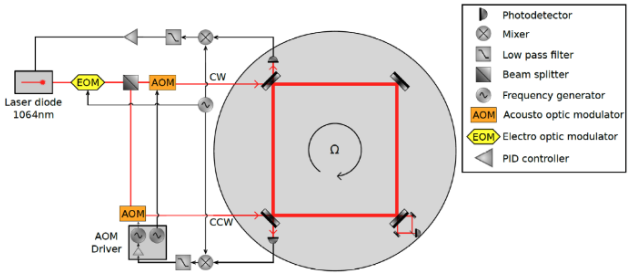


FIG. 1: Schematic of the experimental setup for the passive ring-laser gyroscope. External laser light couples into the square ring cavity. The system utilizes a Pound-Drever-Hall locking scheme to maintain resonance, while a motorized rotation stage provides stable angular velocities for the scale-factor calibration [1].

II. PRE-LAB TASKS

A. Gyroscope Orientations & Maximum Speed

Modern gyroscopes in electronic devices can be used for 3D motion tracking, gaming and navigation among other applications [1] . For the pre-lab tasks, we used an iPad from the year 2022 to analyze its built-in gyroscope using the Phyphox App [3] .

We first started by determining the axis as well as the relative orientation direction ("positive" and "negative") by turning the iPad while measuring the rotation speed using the gyroscope function of the Phyphox app. The result is depicted in Figure 2 .

Given the known axes, we used several methods to try to rotate the iPad as fast as possible. We used a swivel stool, which has the ultimate disadvantage that at high rotation speeds, the iPad will be pushed off the stool if not attached properly. Another method was to flip the

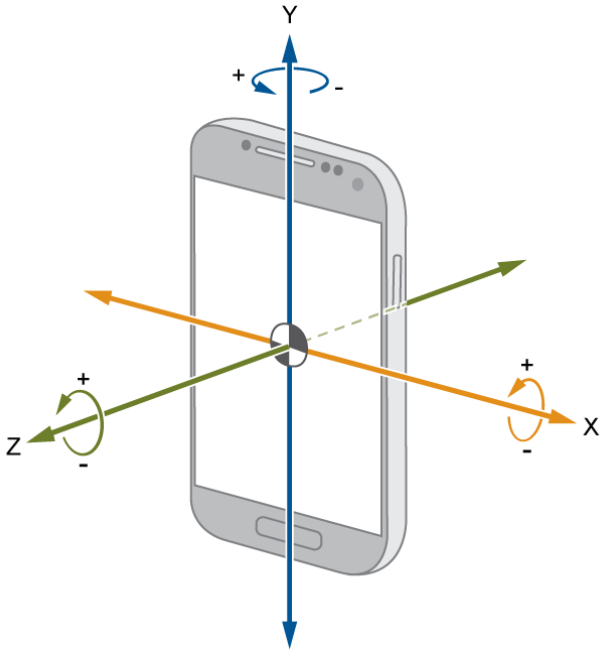


FIG. 2: Axes and relative orientations of the three gyroscopes of the used iPad. The axis associated with the long side of the iPad is the y-axis, while the short axis corresponds to the x-axis. The orthogonal axis pointing out / into the screen is the z-axis. The circular arrows together with the $+$ / $-$ indicate positive / negative rotations. The schematic is taken from [2] .

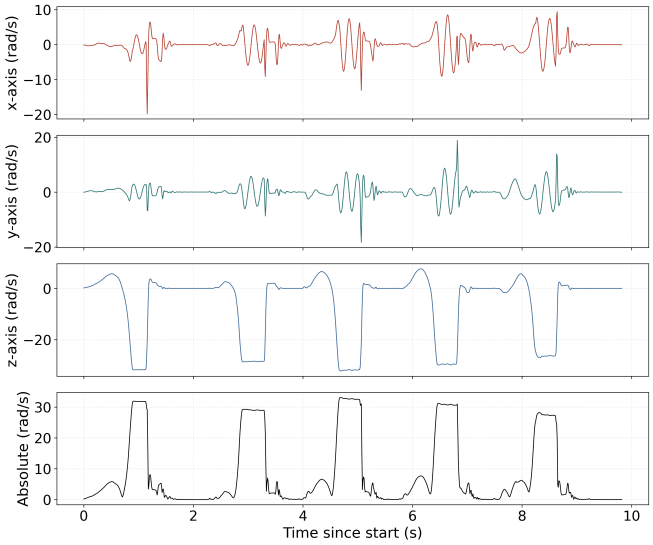


FIG. 3: Time evolution of the measured angular-velocity components during repeated rotation pulses. The upper three panels show the angular velocity along the x-, y-, and z-axes, while the bottom panel shows the corresponding absolute angular velocity (square root of the squared rotation speeds of all three axes in sum). The motion is dominated by the z-component, which exhibits a sequence of nearly rectangular plateaus at about $30\,\mathrm{rad\,s^{-1}}$, whereas the transverse x- and y-components remain comparatively small and contain mainly short oscillatory transients during the switching phases.

iPad in the air above a soft surface to ensure protection. The latter exceeded the top rotation speed of the stool version by more than 50%. The time-series of the iPad flip in mid-air is shown in Figure 3 .

Figure 3 shows all three axes of the iPad's gyroscopes with the highest rotation speed happening around the z-axis, slightly above $30\,\mathrm{rad\,s^{-1}}$. The rotation speed as a function of time has a rectangular-like shape, which can be explained by a fast start of rotation (stark slope at the beginning), followed by spinning in mid-air, where the rotation speed stays nearly constant, then a sharp rise as the iPad hits the soft surface again. Any residual rotations between the rectangular structures are artifacts from picking up the iPad again. The nearly constant rotation speed at the plateaus is likely because spinning around the z-axis exposes the least area to the rotation direction, thereby having the lowest drag.

The main reason why we used the z-axis for spinning is the ease of achieving relatively high rotation speeds while ensuring safety for the electronics. The residual non-zero rotation speeds around both other axes (see top two panels in Figure 3) are an artifact of spinning the iPad with bare hands, which always induces some perpendicular instability / rotations.

B. Allan-Deviation

We recorded a one-hour gyroscope time series using the timed-measurement feature of the Phyphox app, shown in Figure 18 in Appendix A . The iPad was placed on a soft mat in the attic, and the measurement was done with a sampling interval of $\Delta t = 1.0073 \times 10^{-2}\,\mathrm{s}$, corresponding to a sampling rate of 99.3 Hz.

We evaluated the data along all three sensor axes in order to characterize both the spectral noise content and the time-domain stability of the iPad's gyroscopes. Figure 4 shows the corresponding Welch noise spectra on a log-log scale [1] .

All three axes exhibit a broadband noise floor with only a small number of narrow-band features. The most pronounced line appears on the x axis at 18.10 Hz. We further observe weaker peaks at 27.08 Hz on the y axis and at 45.60 Hz on the z axis. Apart from these isolated spectral lines, the spectra do not show strong low-frequency resonances. The data, therefore, indicate that

[1] A Welch noise spectrum is FFT based. The advantage of the Welch-method is that it yields more robust and less noisy power spectral densities.

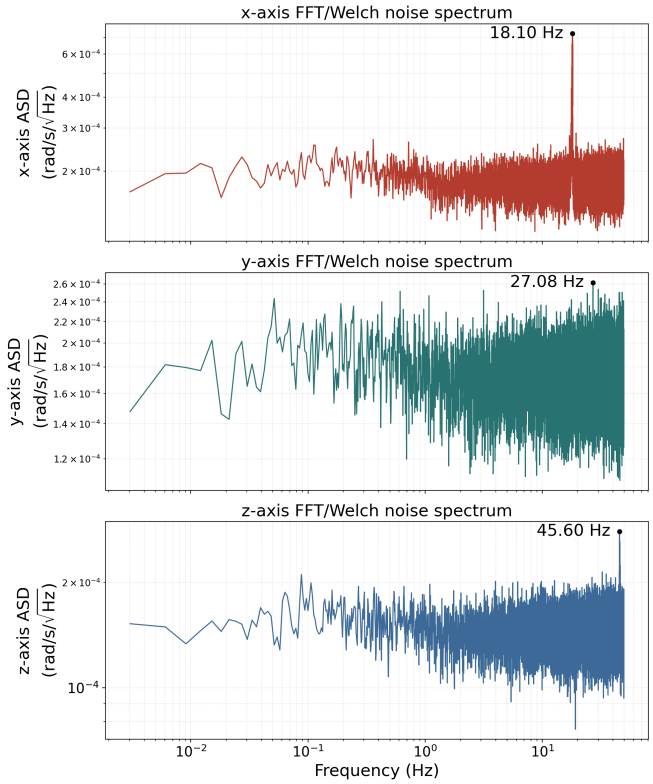


FIG. 4: Log-log Welch noise spectra (FFT-based) of the iPad gyroscope for the x, y, and z axes. The spectra show a broadband noise background together with weak narrow-band peaks at 18.10 Hz (x axis), 27.08 Hz (y axis), and 45.60 Hz (z axis).

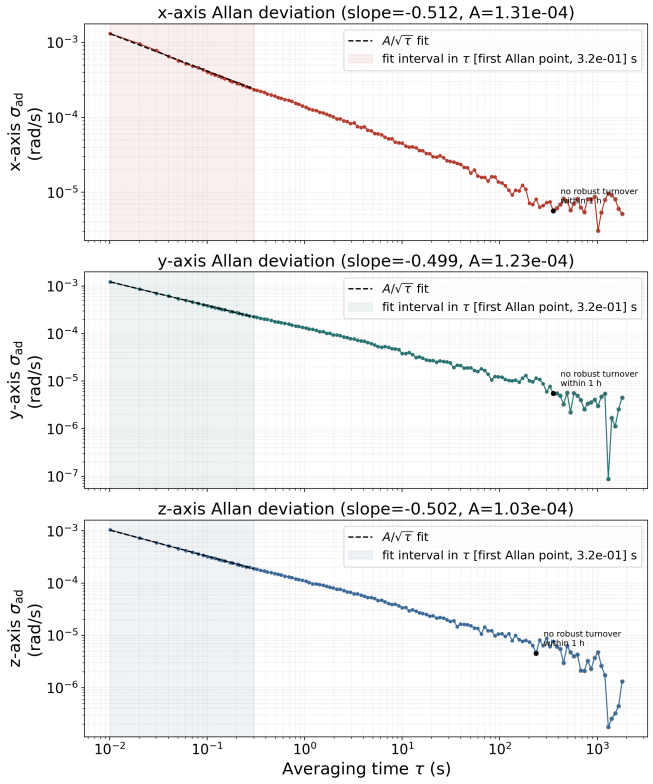


FIG. 5: Allan deviation $\sigma_{\mathrm{ad}}(\tau)$ of the iPad gyroscope for the three sensor axes. The dashed lines denote fits of the form $\sigma_{\mathrm{ad}}(\tau) = A/\sqrt{\tau}$ in the interval from the first Allan point to $\tau = 10^{-0.5}$ s. The measured slopes remain close to $-1/2$, which identifies a white-noise-limited regime and is further discussed in Section II B .

broadband stochastic noise dominates the measurement, while a few narrow-band disturbances likely arise from environmental impacts.

We next quantified the time-domain stability with the Allan deviation $\sigma_{\mathrm{ad}}(\tau)$ [1], as shown in Figure 5. Over the accessible averaging times, all three axes follow the expected $\tau^{-1/2}$ scaling at short and intermediate averaging times τ. We therefore identify this region as the white-noise-limited regime. Strictly speaking, for a micro-electro-mechanical system (MEMS) gyroscope, this regime corresponds to white rate noise or angle random walk rather than photon shot noise; however, the same functional form [1]

$$\sigma_{\mathrm{ad}}(\tau) = \frac{A}{\sqrt{\tau}} \tag{1}$$

still defines the relevant sensitivity coefficient A.

To extract A, we fitted the Allan curves from the first available Allan point up to $\tau = 10^{-0.5}\,\mathrm{s} = 3.16 \times 10^{-1}\,\mathrm{s}$. This specific region was chosen because a "slight" positive systematic deviation is visible if the linear fit is increased further. The reason for this is unknown.

The maximum averaging time that was chosen is half the total measurement time $T = 3600\,\mathrm{s}$ [1] . The resulting slopes are $-0.512(3)$ (x), $-0.499(1)$ (y), and $-0.502(2)$ (z), which confirms the expected $\tau^{-1/2}$ dependence. We further obtain

$$\begin{aligned} A_x &= (1.312 \pm 0.006) \times 10^{-4}\,\mathrm{rad}/\sqrt{\mathrm{s}} \\ A_y &= (1.229 \pm 0.001) \times 10^{-4}\,\mathrm{rad}/\sqrt{\mathrm{s}} \\ A_z &= (1.035 \pm 0.002) \times 10^{-4}\,\mathrm{rad}/\sqrt{\mathrm{s}} \end{aligned} \tag{2}$$

The z axis, therefore, provides the best short-term sensitivity.

The Allan curves do not show a statistically robust minimum within the one-hour measurement. Instead, $\sigma_{\mathrm{ad}}(\tau)$ continues to decrease over the reliably sampled range, and the apparent fluctuations at the largest τ arise from poor statistics because only a few blocks contribute there. We therefore do not identify a reliable turnover time at which slow drift dominates over stability. In other words, this dataset remains white-noise limited over the experimentally accessible averaging times.

For comparison, the Earth's rotation rate is $\Omega_\oplus = 7.29 \times 10^{-5}\,\mathrm{rad/s}$ [4]. Under the white-noise model, the Allan deviation reaches this scale after approximately 3.24 s on the x axis, 2.84 s on the y axis, and 2.01 s on the z axis. This result suggests that, in principle, the gyroscope sensitivity is already sufficient on a few-seconds' averaging scale to measure the earth's rotation speed.

C. Earth Rotation Rate

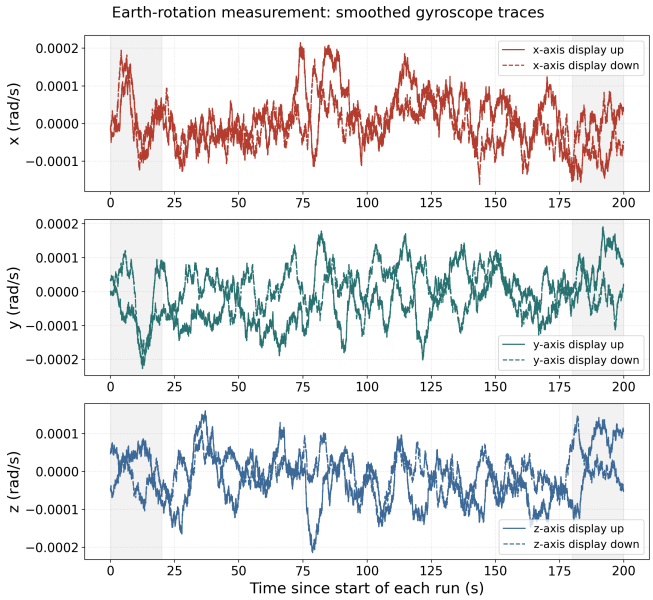


FIG. 6: Gyroscope time series for the unflipped and flipped iPad orientations. The traces show the measured angular-velocity signals before and after the 180° flip around the x-axis. These two signals form the basis for the temporal averages used to extract the Earth-rotation estimate.

To estimate the Earth's rotation rate, we recorded the gyroscope signal of the iPad while the device lay flat on the floor and then flipped it once by 180° about the x-axis. This procedure should reverse the sign of the Earth-rotation projection on the iPad normal, while a static gyroscope offset should remain unchanged. We therefore extracted the rotation signal from the half-difference of the temporal averages before and after the flip.

We performed the analysis for Bonn, Germany, at latitude $\phi = 50.7335°$ [5]. The expected Earth rotation rate is [4]

$$\Omega_\oplus = 7.29 \times 10^{-5}\,\mathrm{rad/s} \tag{3}$$

For a phone lying flat, the screen-normal axis should measure the vertical projection

$$\Omega_\oplus \sin\phi = 5.65 \times 10^{-5}\,\mathrm{rad/s} \tag{4}$$

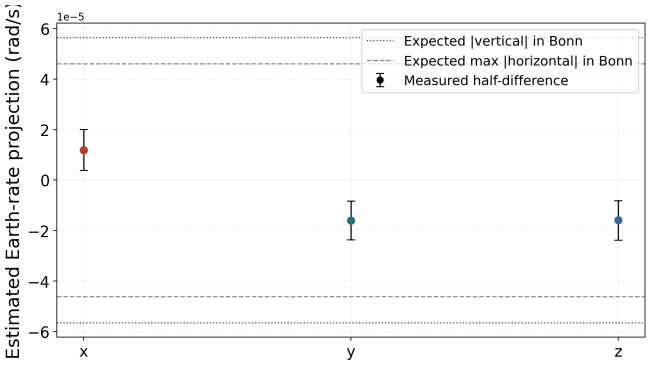


FIG. 7: Axis-resolved Earth-rotation estimates obtained from the half-difference of the temporal averages before and after a flip around the x-axis. The error bars correspond to the Allan-deviation statistical uncertainty. In addition, a dashed line shows the value of the maximal horizontal earth rotation speed in Bonn and a dotted line the expected vertical earth rotation speed in Bonn. The measured z-axis value does not reproduce the expected vertical Earth-rotation signal for Bonn.

whereas the in-plane axes can at most contain the horizontal projection

$$\Omega_\oplus \cos\phi = 4.62 \times 10^{-5}\,\mathrm{rad/s} \tag{5}$$

depending on the azimuthal orientation of the phone [2] . We then determined the axis-resolved Earth-rotation estimate from

$$\hat{\Omega}_i = \frac{\bar{\omega}_{i,\mathrm{pre}} - \bar{\omega}_{i,\mathrm{post}}}{2} \tag{6}$$

Guided by the Allan-deviation analysis of Section II B, we assigned the statistical uncertainty from the Allan deviation at the corresponding averaging times. This procedure yields

$$\hat{\Omega}_x = 1.24(72) \times 10^{-5}\,\mathrm{rad\,s^{-1}}$$

$$\hat{\Omega}_y = -0.81(67) \times 10^{-5}\,\mathrm{rad\,s^{-1}}$$

$$\hat{\Omega}_z = -0.15(58) \times 10^{-5}\,\mathrm{rad\,s^{-1}}$$

Figure 7 summarizes the extracted axis-resolved estimates. The z axis is the most relevant channel for an iPad lying flat because it contains the vertical component of the Earth's rotation vector. However, the measured value

[2] We note that the exact orientation was not properly measured during the measurement.

on this axis remains consistent with zero within the assigned uncertainty and lies 1 order of magnitude below the expected Bonn value of 5.65×10^{-5} rad/s. The x axis shows the largest signal, but it still reaches only about one quarter of the maximum possible horizontal projection. The y value is off by a factor of 5.7.

We therefore conclude that this dataset does not provide a convincing direct measurement of the Earth's rotation. The discrepancy could arise from too few flips, as only one flip was used. Gathering more data could, on average, give a more accurate estimate of the earth's rotation rate.

III. OPERATION OF THE GYROSCOPE

A. Free spectral range and perimeter

To determine the Free Spectral Range (FSR) of the cavity, a Pound-Drever-Hall (PDH) reflection scan [1] was performed. The peak positions for three consecutive resonances, including their respective sidebands, are detailed in Table I.

Since the oscilloscope trace is recorded in the time domain, a calibration factor Cal_i is required to convert the measured time intervals into frequency units. This factor is derived from the known sideband modulation $\Omega = 20\,\mathrm{MHz}$ (representing the $\pm 10\,\mathrm{MHz}$ spacing) and the observed time span $\Delta t_{\mathrm{total},i}$ between the lower and upper sidebands:

$$\mathrm{Cal}_i = \frac{\Omega}{\Delta t_{\mathrm{total},i}} \tag{7}$$

The resulting calibration factors were found to be $1.86(7)\,\mathrm{MHz\,ms^{-1}}$ for the first resonance and $3.2(2)\,\mathrm{MHz\,ms^{-1}}$ for the subsequent two.

The FSR between consecutive longitudinal modes was then calculated using the average calibration factor $\overline{\mathrm{Cal}}$ and the time difference between carrier peaks $\Delta t_{\mathrm{carrier}}$:

$$\delta\nu_{\mathrm{FSR}} = \overline{\mathrm{Cal}} \cdot \Delta t_{\mathrm{carrier}} \tag{8}$$

This yielded FSR values of $281(13)\,\mathrm{MHz}$ and $317(15)\,\mathrm{MHz}$ for the intervals between resonances 1–2 and 2–3, respectively, resulting in a mean value of $\langle \delta\nu_{\mathrm{FSR}} \rangle = 300(10)\,\mathrm{MHz}$ [3]

Finally, the cavity perimeter P was determined using the relation $P = c/\langle \delta\nu_{\mathrm{FSR}} \rangle$, where c is the speed of light. The measured perimeter of $100(3)\,\mathrm{cm}$ is in agreement with the expected value given in the script of $99.0(5)\,\mathrm{cm}$ [1], deviating by $0.37\,\sigma$.

[3] We note that we are not aware of any theoretical value for the given cavity to compare the found value to, except $\delta\nu_{\mathrm{FSR}} = c/P$, but this seems circular to us since we use this formula to calculate the perimeter, which we already compare to the value from the script [1] .

TABLE I: Peak positions from PDH cavity reflection scan.

Resonance	Lower SB	Carrier	Upper SB
Resonance 1	−106.2(3) ms	−101.3(3) ms	−95.5(3) ms
Resonance 2	6.6(3) ms	8.9(3) ms	12.8(3) ms
Resonance 3	104.1(3) ms	107.3(3) ms	110.3(3) ms

B. PDH Error Signal and Slope Estimation

To characterize the frequency discrimination of the locking system, an error signal was generated for Arm A by mixing the photodiode signal with the electro-optical modulator (EOM) modulation frequency. A zoomed-in scan of the resulting Pound-Drever-Hall (PDH) error signal is presented in Figure 8.

A linear fit was applied to the central zero-crossing of the error signal (from 0.0024 s to 0.0025 s) to estimate the discriminator slope, yielding a value of $1.27\,\mathrm{kV\,s^{-1}}$. However, a reliable quantitative estimation of the slope proves difficult in this setup. The signal-to-noise ratio at this scale is insufficient for a high-precision linear fit, as high-frequency noise and residual amplitude modulation (RAM) distort the steepness of the slope. Consequently, while the fit provides a qualitative measure of the signal's polarity and center, it does not represent a robust calibration for the frequency-to-voltage conversion.

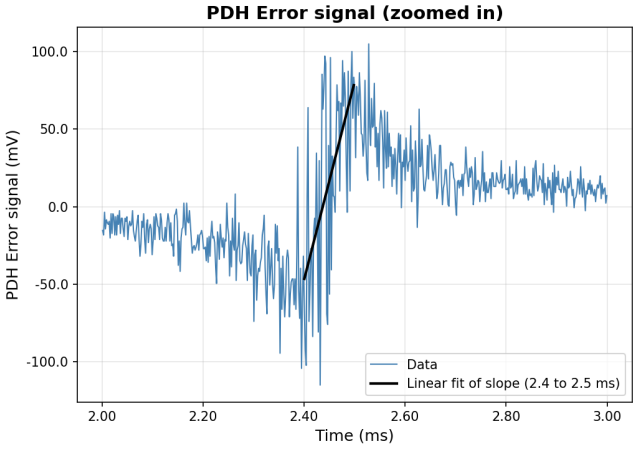


FIG. 8: Pound-Drever-Hall (PDH) error signal with a linear fit applied to the central zero-crossing. The slope of the fit, estimated at $1.27 \times 10^3\,\mathrm{V/s}$, represents the frequency discriminator gain. The high noise level and local fluctuations in the signal make a precise linear estimation challenging.

C. Transversal Mode Characterization

Once the cavity was successfully locked, the transmission was recorded using a camera to analyze the spatial profile of the resonant field. The resulting intensity pattern is shown in Figure 9.

In the context of the Hermite-Gaussian modes ($TEM_{m,n}$) discussed in Section 3.3.1 of [1], the observed transmission shows a symmetric, circular intensity distribution. The absence of observable nodes along either the horizontal or vertical axes indicates that the laser is successfully coupled to the fundamental mode ($\mathrm{TEM}_{0,0}$). Ensuring the excitation of this Gaussian transverse mode is critical for the stability of the gyroscope, as higher-order modes would introduce different Gouy phase shifts and lead to mode competition, potentially degrading the sensitivity of the Sagnac frequency measurement.

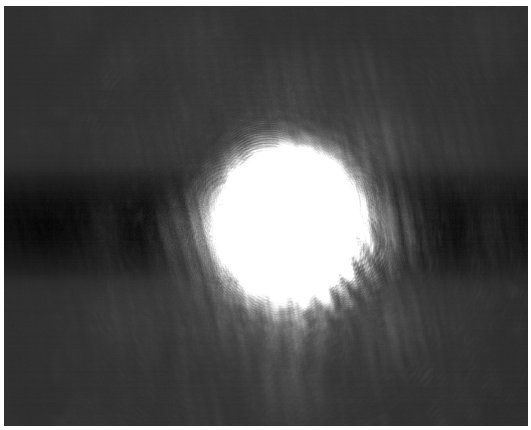


FIG. 9: Transmitted intensity profile of the ring cavity under locked conditions, captured via a CMOS camera. The symmetric, circular distribution without observable horizontal or vertical nodes confirms successful coupling to the fundamental Gaussian mode ($\mathrm{TEM}_{0,0}$). This spatial mode purity is essential for minimizing frequency shifts in the Sagnac signal.

D. Cavity Ringdown

1. Experimental Methodology

To characterize the losses and the finesse of the optical cavity, we performed a cavity ringdown measurement as described in [1] . We locked one laser beam to the cavity resonance, while the transmission of the counter-propagating beam was monitored using a fiber-coupled photodiode. The oscilloscope was configured to trigger on the transmission signal at a level between the locked intensity and the background noise. Upon rapidly switching off the laser beam, the stored energy in the cavity leaks out, resulting in an exponential decay of the transmitted intensity.

2. Results and Analysis

The decay time constant τ was extracted by fitting an exponential function of the form $I(t) = I_0 e^{-t/\tau}$ to the recorded data. Despite how bad the fits look in the plots the reduced $\chi 2$ was almost 1. for all fits was From this decay time, we determined the reflectivity of the cavity mirrors R (assuming the same reflectivity for all four mirrors) and the finesse F using the relations: [1]

$$R^2 = \exp\left(-\frac{1}{\tau \cdot \delta\nu_{\mathrm{FSR}}}\right) \tag{9}$$

$$F = \frac{\pi \cdot R}{1 - R^2} \tag{10}$$

We summarize the results for twelve consecutive measurements in Table II.

The measurements exhibit significant variance, with the calculated finesse values ranging from approximately $13.9(15)\times10^3$ to $342(3)\times10^3$. This discrepancy is largely attributed to poor data quality likely caused by a coarse time resolution setting on the picoscope during data acquisition. As a result, the exponential decay was very noisy. Figure 10 illustrates a particularly noisy trace where the fit seems unreliable, whereas Figure 11 shows a relatively better fit despite the underlying noise issues.

The mean decay time was found to be 78(27) µs, corresponding to an average finesse of $74(26)\times10^3$. To improve the precision of these results in future measurements, a significantly finer time resolution could be employed to properly resolve the decay curve and minimize the statistical uncertainty in the fit parameters.

The statistical uncertainties of the fit parameter τ were orders of magnitude smaller than the fluctuations we found while trying out different fit windows. Thus, we assigned a 10 % relative uncertainty to capture this systematic uncertainty. We acknowledge that a more quantitative analysis of the systematic uncertainty could have been done by varying the fit windows and observing the changes in the fit parameter. However, this would have been beyond the scope of this report.

IV. SCALE-FACTOR CALIBRATION

A. Measurement of the Sagnac-Beat

For an optical ring cavity whose area vector $\vec{A}$ is aligned with the rotation vector $\vec{\Omega}$, the Sagnac beat frequency $\delta\nu$ is given by

$$\delta\nu = \frac{4\vec{A}\cdot\vec{\Omega}}{\lambda P} \equiv K\Omega, \tag{11}$$

where $A \equiv |\vec{A}|$ is the enclosed area, P is the cavity perimeter, λ is the laser wavelength, $\Omega \equiv |\vec{\Omega}|$ is the rotation speed and K is the gyroscope's scale factor [1].

TABLE II: Cavity ringdown results using the exact physics relations. Individual errors are propagated from τ (10%) and FSR (300 ± 10 MHz). Mean values include the Standard Error of the Mean (SEM).

File #	τ (µs)	F	R	Loss (ppm)
1	92.46	$87.1(92) \times 10^3$	0.999 964(4)	36.1
2	167.58	$157.9(166) \times 10^3$	0.999 980(2)	19.9
3	61.38	$57.9(61) \times 10^3$	0.999 946(6)	54.3
4	342.22	$322.5(340) \times 10^3$	0.999 990(1)	9.7
5	43.38	$40.9(43) \times 10^3$	0.999 923(8)	76.8
6	32.40	$30.5(32) \times 10^3$	0.999 897(11)	102.9
7	39.09	$36.8(39) \times 10^3$	0.999 915(9)	85.3
8	17.17	$16.2(17) \times 10^3$	0.999 806(20)	194.2
9	88.12	$83.1(88) \times 10^3$	0.999 962(4)	37.8
10	15.06	$14.2(15) \times 10^3$	0.999 779(23)	221.3
11	23.49	$22.1(23) \times 10^3$	0.999 858(15)	141.9
12	14.75	$13.9(15) \times 10^3$	0.999 774(24)	226.0
Mean	78(27)	$74(26) \times 10^3$	0.999 900(2)	101(22)

For the square cavity used here, $A = L^2$ and $P = 4L$, Equation (11) can further be simplified to

$$K_{\text{th}} = \frac{4A}{\lambda P} = \frac{P}{4\lambda}. \tag{12}$$

With a literature perimeter of $P = 0.990(5)\,\text{m}$ and a wavelength of $\lambda = 1064\,\text{nm}$ [1], the predicted scale factor K_{th} is

$$K_{\text{th}} = 232\,613(1175)\,\text{Hz}\,\text{rad}^{-1}\,\text{s}^{-1}. \tag{13}$$

We performed the measurement by using a power supply to rotate the cavity setup and simultaneously measure the Sagnac-beat $\delta\nu$. The Sagnac-beat corresponds to the signal difference of the two acousto optical modulators (AOMs). To ensure a valid measurement of $\delta\nu$, we locked the two counterpropagating beams as described in Section 5.2 of [1] .

B. Data Selection

The recorded files contain the lock-monitor voltage $V_{\text{lock}}(t)$, the beat-frequency trace $\delta\nu(t)$, and the encoder-frequency information required for the analysis. We measured in both the positive rotation direction as well as the negative one. In addition, we use the lock-monitor signal as the sole selection criterion. The medians of the lock-monitor voltage separate into four distinct branches with centroids at 0.017 V, 0.043 V, 0.096 V and 0.123 V. Visual inspection of the corresponding beat-frequency traces confirms that only the lowest branch is clearly in an unlocked state. Through visual inspection, we found that imposing a cut at

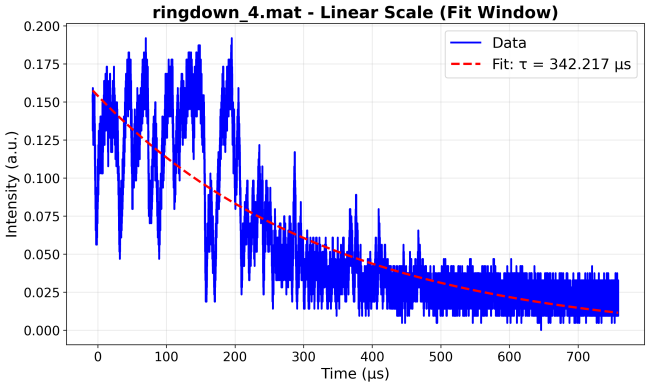


FIG. 10: Example of a ringdown fit with significant noise, leading to an artificially high finesse estimate.

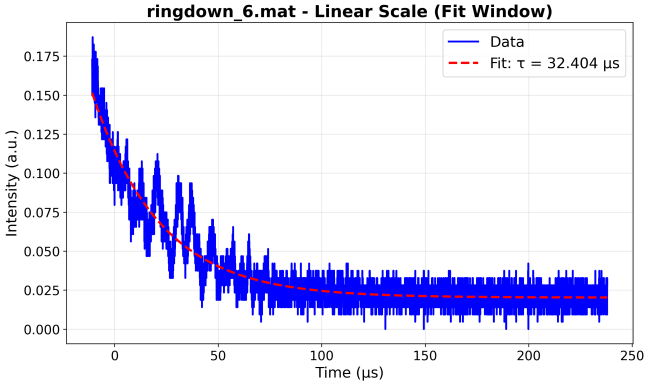


FIG. 11: A comparatively cleaner ringdown trace, though still limited by high noise of the acquisition.

$$\text{median}(V_{\text{lock}}) \geq 0.03\,\text{V}, \tag{14}$$

separates all clearly unlocked Sagnac-beat files from clearly locked ones. The cut value corresponds to the midpoint between the deep-unlock branch and the next higher branch.

Figures 19 and 20 in Appendix B show the resulting acceptance and rejection of the files. The same threshold is applied to both the positive and negative direction files so that all data is selected by a single, symmetric physical criterion.

For every retained file, the Sagnac observable is defined as the arithmetic mean of the instantaneous beat-frequency values. At a fixed motor voltage, these file-wise means are averaged, and the standard error of the mean (SEM) is taken as the statistical uncertainty (see Appendix C).

C. Encoder Calibration

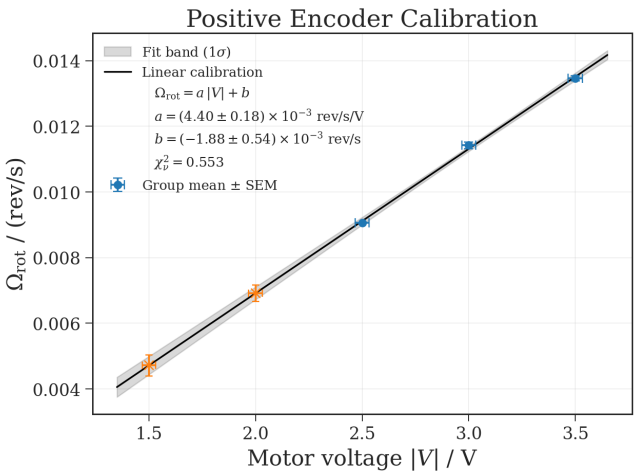


FIG. 12: Encoder calibration used to obtain the low-voltage angular speeds for the positive branch. The direct encoder points come from retained files only, the horizontal error bars encode the voltage-source uncertainty, and the crosses denote the extrapolated low-voltage values.

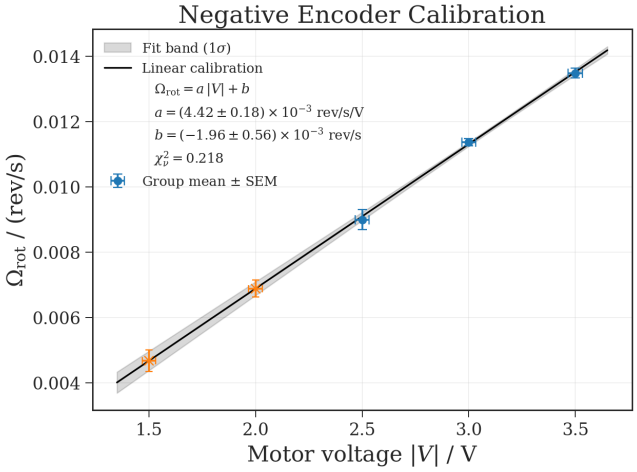


FIG. 13: Corresponding encoder calibration for the negative branch. The calibration is performed separately for the two directions before both branches are combined in the signed fit.

The rotary encoder provides the primary measurement of the table's angular speed. The saved encoder frequency measures complete cycles of one encoder signal. Since one cycle corresponds to four quadrature edge counts, the mechanical rotation frequency is

$$f_{\mathrm{rot}} = \frac{f_{\mathrm{enc}}}{2500}, \tag{15}$$

and the rotation speed Ω entering the fit is

$$\Omega = 2\pi \frac{f_{\mathrm{enc}}}{2500}. \tag{16}$$

Because direct encoder information was not available for the lower voltage files, the 1.5 V and 2.0 V points are extrapolated separately for the positive and negative directions using a linear calibration based on the higher-voltage readings. We assign each voltage setting an uncertainty of

$$\sigma_V = 0.001\,|V| + 0.03\,\mathrm{V}, \tag{17}$$

coming from the inherent measurement uncertainty of the power supply used [6] . The grouped encoder calibration points therefore carry horizontal error bars from Equation (17) and vertical error bars from the standard error of the mean encoder rate.

It should be noted that we used a resolution of 100 mV on the power supply screen, which means fluctuations on higher orders, separate from the inherent uncertainty given by the manufacturer [6], were not resolvable. In addition, the fact that voltages are usually rounded by the power supply to match the set resolution, the given total uncertainty in Equation (17) in the motor voltage might be an underestimation. The primary impact would be on the calibration curve and the extrapolated values for rotation speeds, systematically altering everything that is based on rotation speeds (Sagnac-effect and with this also the earth's rotation rate) .

D. Signed Data Points and Fitting Procedure

To extract the scale factor K, we model the Sagnac beat frequency as a linear function of the applied angular velocity which is motivated by the theoretical relation given in Equation (11). The physically motivated baseline model is

$$\delta\nu = K\Omega, \tag{18}$$

because the Sagnac effect predicts a beat frequency proportional to the rotation rate. In this model, the slope directly yields the gyroscope scale factor K.

In addition, we also consider the extended model

$$\delta\nu = K\Omega + b, \tag{19}$$

which allows for a non-zero intercept b, which is more realistic as even at 0 rotation speed, the earth, for instance, will rotate and cause a non-zero Sagnac-beat. This fact will become important in Section VII C.

The resulting signed data points entering the combined fit are listed in Table III. Because the x-axis uncertainties

TABLE III: Signed data points used in the combined scale-factor fit. The rotation speeds Ω and Sagnac-beat $\delta\nu$ carry the sign of the applied table rotation, while the uncertainties remain positive. N indicates the number of retained files. Each voltage U point has an associated error of $0.1\% \times U + 30\,\mathrm{mV}$ coming from the uncertainty of the power supply used [6].

Point	Signed Ω (mrad/s)	Origin of Ω	Signed $\delta\nu$ (Hz)	N
−3.5 V	−84.68(95)	encoder	−15 364(657)	18
−3.0 V	−71.40(67)	encoder	−14 906(1261)	13
−2.5 V	−56.52(196)	encoder	−15 121(809)	13
−2.0 V	−43.21(300)	calibration	−10 010(342)	6
−1.5 V	−29.33(361)	calibration	−6118(396)	15
1.5 V	31.53(272)	calibration	10 928(365)	34
2.0 V	44.82(223)	calibration	11 220(871)	20
2.5 V	56.97(183)	encoder	15 954(785)	14
3.0 V	71.78(65)	encoder	18 166(779)	16
3.5 V	84.59(48)	encoder	20 354(893)	16

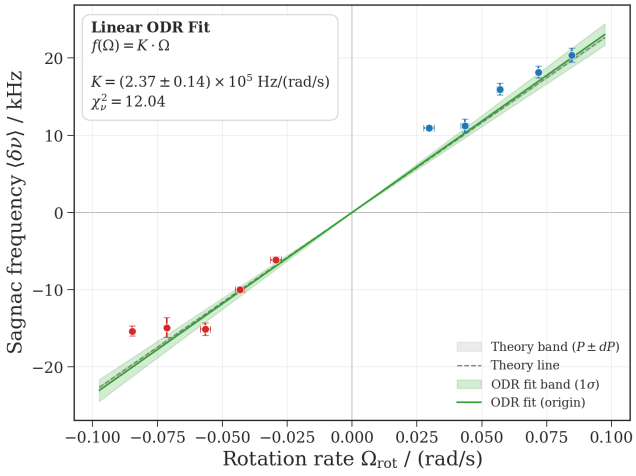


FIG. 14: Signed scale-factor fit constrained to pass through the origin. Positive and negative rotation directions enter simultaneously. The gray band shows the geometrical expectation from Equation (13), and the green band gives the 1σ fit uncertainty band.

are non-negligible due to the extrapolation procedure, both fit models (Equations 18 and 19) are evaluated using orthogonal distance regression (ODR). This ensures that uncertainties on both axes are simultaneously weighted in the determination of the fit parameters.

E. Results and Discussion

The signed fit through the origin (Figure 14) yields

$$K = 236\,588(14355)\,\mathrm{Hz\,rad^{-1}\,s^{-1}}, \tag{20}$$

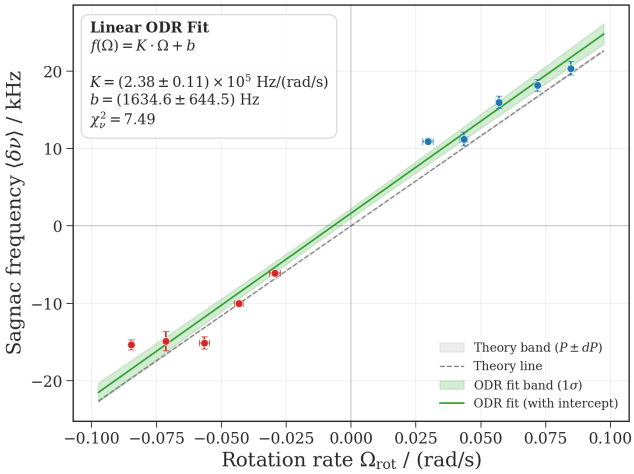


FIG. 15: Signed scale-factor fit with a free intercept. In the signed representation, the intercept corresponds to a Sagnac shift that does not change sign with the applied table rotation, serving as the natural candidate for a direction-independent terrestrial contribution.

TABLE IV: Summary of the two signed-fit models. The slope K is the gyroscope scale factor and the intercept b in the second fit is the candidate direction-independent beat offset. The values are quoted with integer precision.

Model	K (Hz s/rad)	b (Hz)	χ^2_ν
$\delta\nu = K\Omega$	236 588(14355)	—	12.0
$\delta\nu = K\Omega + b$	237 888(11278)	1635(644)	7.5
Theory	232 613(1175)	—	—

whereas the signed free-intercept fit (see Figure 15) returns

$$\begin{aligned} K &= 237\,888(11278)\,\mathrm{Hz\,rad^{-1}\,s^{-1}}, \\ b &= 1635(644)\,\mathrm{Hz}. \end{aligned} \tag{21}$$

As summarized in Table IV, both evaluated slopes agree with the theoretical expectation from Equation (13) within their 1σ interval.

Additionally, when we use the in Section III A experimentally determined perimeter value of $P = 100(3)\,\mathrm{cm}$ and Equation (12) , the experimental reference value for the scaling factor is

$$K_{\mathrm{exp}} = 234\,962(7048) \tag{22}$$

which also agrees within its 1σ interval with both fit values listed in Table IV (and the fit values agree within their respective 1σ interval with K_{exp}). Therefore, we found no statistically significant deviation from the theoretical value K_{th} and all experimentally obtained scaling factor values.

The slope extracted from the combined positive and negative data remains stable when the intercept is released, and both values of K are consistent with Equation (13) as well as with each other within their 1σ uncertainty band indicating statistical compatibility. This agreement shows that the scale factor determination is robust against the transition from a positive-only to a signed two-branch analysis.

In an ideal signed linear picture, the positive and negative values should have identical absolute size, differing only by sign. A simple and direct comparison is obtained by pairing the positive and negative Sagnac shifts listed in Table III at equal absolute motor voltage and comparing only their magnitudes. We see, that the data do not satisfy the symmetry around the y-axis uniformly. The closest agreement is found at 2.5 V, where $|\delta\nu(+)| = 15\,954$ Hz and $|\delta\nu(-)| = 15\,121$ Hz, corresponding to an absolute mismatch of 833 Hz or only 5.4% relative to their mean magnitude. The 2.0 V pair is also comparatively close, with $|\delta\nu(+)| = 11\,220$ Hz and $|\delta\nu(-)| = 10\,010$ Hz, i.e. a mismatch of 1210 Hz or 11.4%. By contrast, the disagreement becomes pronounced at 3.0 V and 3.5 V, where the positive magnitudes exceed the negative ones by 3260 Hz (19.7%) and 4990 Hz (27.9%), respectively. The strongest asymmetry appears already at 1.5 V, where $|\delta\nu(+)| = 10\,928$ Hz and $|\delta\nu(-)| = 6118$ Hz, giving a difference of 4811 Hz, or 56.4%. Thus, only the 2.0 V and especially the 2.5 V pairs come close to the symmetry expected from the signed linear model, whereas the 1.5 V, 3.0 V, and 3.5 V pairs are clearly unequal in magnitude. As a result, the reduced chi-square of $\chi^2_\nu = 12.0$ for the origin-constrained fit (see Figure 14) does not merely reflect random point-to-point fluctuations, but indicates a clear systematic mismatch between the model $\delta\nu = K\Omega$ or $\delta\nu = K\Omega + b$ and the data. This conclusion is also borne out by visual inspection. The three lowest negative-voltage points and the first positive-voltage point do not follow a steadily rising linear trend through the origin. Instead, they form two approximately flat groups: the lowest positive point is mutually consistent within its uncertainty with the 2.0 V point, and the three lowest negative points are likewise mutually consistent within their uncertainties. In other words, over this low-rotation range the measured beat frequency changes much less strongly with Ω than both linear models predict.

Allowing for a free intercept reduces the reduced chi-square from 12.0 to 7.5, which shows that part of the systematic mismatch is indeed absorbed by a direction-independent offset term. Within the class of linear models considered here, this makes the free-intercept fit the statistically more appropriate description of the signed data. At the same time, $\chi^2_\nu = 7.5$ is still far above unity which is likely explained by the fact that the model doesn't properly describe the data points, leading to increased sizes of the residuals and therefore, in sum, a larger χ^2_ν value. The difference in χ^2_ν between the origin-bound linear model and the extended linear model directly comes from the one additional degree of freedom in the extended model. This makes it statistically possible to "better" describe the data.

However, due to the asymmetry of the positive and negative direction data points, we conclude that during the measurement, something went wrong. A potential way to improve statistics is to take more data as more mean estimates will yield increasingly higher accuracies.

V. ALLAN DEVIATION

The stability and sensitivity of the gyroscope were evaluated by computing the overlapping Allan deviation, $\sigma_{\mathrm{ad}}(\tau)$, from the rotation-rate measurements. The primary dataset consisted of 200 individual files, each containing 50 ms of signal, for a total effective measurement time of 10 s. Although a 30 s duration was intended, the discrepancy possibly stems from the overhead in the discrete file-writing process or a loss of continuous signal trace.

To create a continuous time series for analysis, the 200 files were concatenated. To ensure this procedure did not introduce spectral artifacts, a boundary-jump analysis was performed across all 199 transitions. The jumps exhibited a mean of approximately zero and a standard deviation comparable to the inherent noise of the within-file signal, suggesting that concatenation did not introduce systematic discontinuities that would bias the resulting Allan deviation.

The processing pipeline involved extracting the Sagnac beat frequency from the photodiode signal using a Hilbert transform to determine the instantaneous frequency. This frequency data was subsequently converted to a rotation rate using the scale factor K found in Section IV. It should be noted, per the tutor's warning, that the frequency-extraction procedure and bandpass ringing can induce non-physical features at short averaging times. Consequently, the identification of a true shot-noise-limited region requires careful scrutiny.

An examination of the Allan deviation curve, shown in Figure 16 , reveals the absence of a clearly extended $\tau^{-1/2}$ region. Instead, the local slope, $\mathrm{d}\log\sigma_{\mathrm{ad}}/\mathrm{d}\log\tau$, oscillates across the accessible range and fails to sustain a value near the theoretical shot-noise expectation of -0.5. This indicates that at a motor voltage of -1.5 V, the measurement is not strictly limited by white photon noise, but is instead limited by other effects.

Given the lack of a single, clean shot-noise window, the data was characterized using two distinct fit regions. The main fit, shown in Figure 16, covers the interval $\tau \in [2.0\,\mathrm{ms}, 5.7\,\mathrm{ms}]$, which represents the longest approximately linear descent visible in the curve. A free-slope linear fit in this region yields a slope of $-2.39(16)$—substantially steeper than the shot-noise expectation. When the slope is constrained to $-1/2$, we obtain a Shot-noise limited sensitivity of $A_{\mathrm{main}} = 0.144\,\mathrm{mrad}/\sqrt{\mathrm{s}}$. This fit achieved a reduced χ^2 of 34,918 showing the not shot

noise limited nature of the allan deviation.

A second, short-τ fit was performed in the window $\tau \in [0.15\,\mathrm{ms}, 0.50\,\mathrm{ms}]$, as illustrated in Figure 17 . In this narrow region, the local slope comes closest to the theoretical value, with a free-slope fit yielding $-0.51(9)$. A constrained fit here results in $A_{\mathrm{short}} = 0.067\,\mathrm{mrad}/\sqrt{\mathrm{s}}$. However, because this region coincides with the short-averaging-time domain flagged as susceptible to Hilbert transform edge effects and filter ringing, these values are reported for completeness and are not considered the primary result. Ultimately, the main fit provides the most robust, albeit conservative, upper bound on the shot-noise sensitivity of the system. This fit achieved a reduced χ^2 of 277 which is still not good but much shows much better agreement than the 34,918 of the other fit.

VI. THEORETICAL SENSITIVITY COMPARISON

The fundamental performance limit of the gyroscope is governed by the quantum shot noise of the detected photons. According to the theoretical model, the sensitivity to a rotation rate Ω for an integration time τ is characterized by the Angle Random Walk (ARW) coefficient A, defined as:

$$A = \frac{1}{4}\frac{c\lambda}{L^2 F}\sqrt{\frac{h\nu}{P_{\mathrm{opt}}}} \tag{23}$$

Using the experimental parameters of $P_{\mathrm{opt}} = 1\,\mu\mathrm{W}$, a cavity perimeter $L = 100(3)\,\mathrm{cm}$, a wavelength $\lambda = 1064\,\mathrm{nm}$, and the measured finesse $F = 74(26) \times 10^3$, the theoretical sensitivity is calculated to be:

$$A_{\mathrm{theory}} = 4.7(17) \times 10^{-10}\,\mathrm{rad}/\sqrt{\mathrm{s}} \tag{24}$$

Comparing this value to the experimental sensitivity extracted from the Allan deviation (Figure 16), which yielded $A_{\mathrm{exp}} = 1.44 \times 10^{-4}\,\mathrm{rad}/\sqrt{\mathrm{s}}$, reveals a discrepancy of approximately five orders of magnitude.

This result indicates that the current measurement at a motor voltage of $-1.5\,\mathrm{V}$ is not shot-noise limited. This is further supported by the free-slope fit of the Allan deviation, which yielded a slope of -2.39 ± 0.16 rather than the -0.5 expected for white photon noise. The observed performance is likely dominated by technical noise sources, including mechanical vibrations, laser relative intensity noise (RIN), and artifacts introduced during the frequency-extraction process, such as bandpass filter ringing and Hilbert transform edge effects.

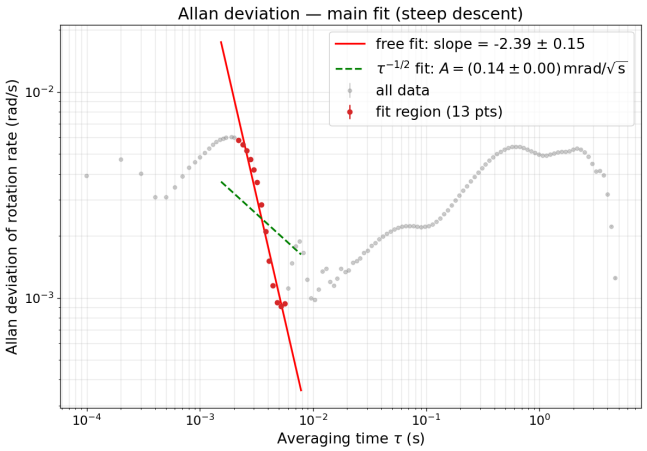


FIG. 16: Allan deviation fit over the main descent region. The data (gray points) show the overlapping Allan deviation computed from 10 s of rotation-rate measurement. The highlighted red points span the steepest descent ($\tau \approx 2\,\mathrm{ms}$ to $5.7\,\mathrm{ms}$). The red solid line shows a free-slope fit (slope -2.39 ± 0.16), while the green dashed line shows a slope constrained to $-1/2$, yielding $A = 0.144\,\mathrm{mrad}/\sqrt{\mathrm{s}}$.

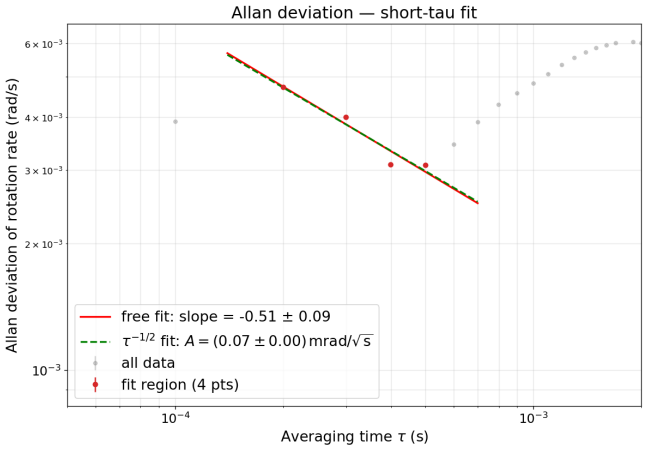


FIG. 17: Allan deviation fit over the short-τ region. Zoom of the window $\tau \in [0.15\,\mathrm{ms}, 0.50\,\mathrm{ms}]$. The highlighted red points form a shallow dip where the slope ($-0.51(9)$) is consistent with shot noise. The constrained green dashed line gives $A = 0.067\,\mathrm{mrad}/\sqrt{\mathrm{s}}$. This region is susceptible to processing artifacts.

VII. EARTH'S ROTATION RATE

A. The Lock-In Effect and Its Threshold

The lock-in effect is universal in weakly coupled oscillators: two oscillators with slightly different natural frequencies synchronize to exactly the same frequency when the coupling exceeds the frequency difference. In the ring-laser gyroscope the two oscillators are the counter-

propagating cavity modes, and their coupling is mirror backscattering—a small fraction of the clockwise mode power is scattered into the counter-clockwise mode and vice versa. Whenever the Sagnac-beat that the rotation would produce falls below the lock-in threshold Ω_L, the two modes lock to the same frequency and the beat disappears entirely [1].

A standard estimate for the lock-in threshold is [1]

$$\Omega_L = \frac{c\lambda^2 r_s}{32\pi A d}, \tag{25}$$

where c is the speed of light, A the enclosed cavity area, d the beam diameter at the mirrors, and r_s the backscattered-power fraction relative to total round-trip loss. The backscattering fraction satisfies the upper bound [1] $r_s^2 \lesssim 4(1-R)$, where $R = R_1R_2R_3R_4$ is the product of all four mirror reflectivities.

Numerical estimate

For the setup used, we use $\lambda = 1064\,\mathrm{nm}$, $A = (1.00(3)\,\mathrm{m}/4)^2 = 0.0625(38)\,\mathrm{m}^2$, $d = 300\,\mu\mathrm{m}$ [1], and $R = 0.999\,900(2)$ (average value used from Table II), giving

$$r_s \lesssim \sqrt{4(1-0.999\,900(2))} = 0.0200(2). \tag{26}$$

Substituting into Equation (25) and propagating the uncertainties from A and r_s yields

$$\Omega_L \lesssim 3.60(22)\times 10^{-3}\,\mathrm{rad\,s^{-1}}. \tag{27}$$

The expected Earth-rotation signal at Bonn ($\phi = 50.7335^\circ$ [5] and $\Omega_\oplus = 7.29\times 10^{-5}\,\mathrm{rad/s}$ [4]) is

$$\Omega_{\oplus,\mathrm{Bonn}} = \Omega_\oplus \sin\phi \approx 56.46\,\mu\mathrm{rad\,s^{-1}}, \tag{28}$$

and the corresponding Sagnac-beat that Earth's rotation alone would produce is $K_{\mathrm{th}}\,\Omega_{\oplus,\mathrm{Bonn}} \approx 0.01\,\mathrm{kHz}$. The ratio of the lock-in threshold to this signal is

$$\frac{\Omega_L}{\Omega_{\oplus,\mathrm{Bonn}}} \approx 64(4). \tag{29}$$

The lock-in threshold exceeds the Earth signal by nearly two orders of magnitude. Any Sagnac beat from the Earth's rotation lies well inside the lock-in region, making direct observation impossible with this device regardless of measurement duration.

B. Measurement Techniques

If the signed free-intercept model in Equation (19) is interpreted literally, the intercept b represents the remaining Sagnac shift that is invariant under sign reversal of the rotation. The corresponding Earth-projected angular velocity is therefore

$$\Omega_{\oplus,\mathrm{fit}} = \frac{b}{K}. \tag{30}$$

The same physical principle can be expressed directly for each pair of opposite rotation directions. At a fixed absolute voltage $|V|$, the subtraction

$$\Delta f(|V|) = \delta\nu_+(|V|) + \delta\nu_-(|V|) \tag{31}$$

removes the dominant table-rotation contribution under the signed convention. The terrestrial rotation rate $\Omega_\oplus$ can then be extracted as

$$\Omega_{\oplus,\mathrm{sub}}(|V|) = \frac{\Delta f(|V|)}{2K}. \tag{32}$$

For the laboratory's latitude in Bonn ($\phi = 50.7335^\circ$ [5]), the expected projection is

$$\Omega_{\oplus,\mathrm{Bonn}} = \Omega_\oplus \sin\phi \approx 56.46\,\mu\mathrm{rad\,s^{-1}}. \tag{33}$$

C. Data Evaluation Procedure

We first evaluate the intercept-based proxy by propagating the full covariance matrix of the free-intercept fit through Equation (30). Because K and b are strongly correlated, we retain the covariance and use it in the propagation.

We then evaluate the direct subtraction method (Equation 32) for each absolute voltage pair. We combine the positive and negative beat frequencies listed in Table III with the slope K from the signed free-intercept fit. The uncertainty of each subtraction point includes both the uncertainty of Δf and the uncertainty of the fitted slope. Finally, we form a weighted mean over the five independent subtraction estimates.

D. Results and Discussion

Evaluating Equation (30) with the parameters from Equation (21) yields

$$\Omega_{\oplus,\mathrm{fit}} = 6871(2701)\,\mu\mathrm{rad\,s^{-1}}. \tag{34}$$

TABLE V: Earth-rotation estimates obtained from the direct subtraction of opposite-direction pairs. For each absolute motor voltage, the table gives the opposite-direction beat-frequency difference Δf and the resulting $\Omega_{\oplus,\mathrm{sub}}$ obtained via Equation (32).

$\lvert V\rvert$	Δf (Hz)	$\Omega_{\oplus,\mathrm{sub}}$ ($\mu\mathrm{rad\,s^{-1}}$)
1.5 V	4811(538)	10 110(1229)
2.0 V	1210(936)	2544(1971)
2.5 V	833(1127)	1751(2371)
3.0 V	3260(1482)	6853(3133)
3.5 V	4990(1108)	10 490(2382)

The intercept-based estimate exceeds the expectation by a factor of roughly 121. The direct subtraction analysis detailed in Table V produces a weighted mean of

$$\Omega_{\oplus,\mathrm{sub}} = 7421(853)\,\mu\mathrm{rad\,s^{-1}}, \tag{35}$$

This secondary estimate exceeds the expected Bonn projection by a factor of roughly 132.

The two extraction methods point to the exact same qualitative conclusion. The intercept-based estimate (Equation 34) and the subtraction-based estimate (Equation 35) agree with one another well within uncertainty, demonstrating that both frameworks are sensitive to the same direction-independent offset structure. However, because both extracted values reside approximately two orders of magnitude above the target signal (see Equation 33), the data clearly do not support a successfully resolved measurement of the Earth's true rotation.

The comparison remains highly instructive because the two techniques parse the available data differently. The signed-fit intercept compresses the full aggregate dataset into one global offset parameter, whereas the pairwise subtraction method tests the cancellation symmetry point-by-point across absolute voltage magnitudes. Their mutual agreement strongly suggests that the excess signal is not merely an artifact tied to a single faulty voltage run. Instead, it points to a systematic offset that robustly survives the sign reversal of the table's motion, potentially indicating measurement faults in the Laser locking as this was one of the most volatile parts of the setup, seemingly randomly locking itself and unlocking inconsistently of outer influences.

VIII. CONCLUSION

We characterized the fundamental properties and rotational sensitivity of a passive ring-laser gyroscope. As detailed in Section III A, we verified the cavity geometry by measuring a free spectral range of 300(10) MHz, yielding a perimeter of 100(3) cm. Furthermore, utilizing the cavity ringdown method (Section III D), we extracted a mean photon decay time of 78(27) µs, corresponding to an average cavity finesse of $74(26) \times 10^3$ and a total round-trip loss of 101(22) ppm.

We established a self-consistent framework to calibrate the scale factor of the passive ring-laser gyroscope by merging positive and negative rotation data. Using the lock-monitor median as a selection criterion allowed us to isolate functional measurement branches of the corresponding Sagnac beat signals. Our signed free-intercept discussed in Section IV model yielded a scale factor of $K_{\mathrm{intercept}} = 237\,888(11278)\,\mathrm{Hz\,rad^{-1}\,s^{-1}}$, which matches the geometrical expectation of $K_{\mathrm{th}} \approx 232\,613(1175)\,\mathrm{Hz\,rad^{-1}\,s^{-1}}$ within its 1σ interval.

Despite this calibration, our analysis proves that the current apparatus cannot resolve the Earth's rotation signal. Extracting the terrestrial Sagnac shift from the global fitted intercept produced $\Omega_{\oplus,\mathrm{fit}} = 6871(2701)\,\mu\mathrm{rad\,s^{-1}}$, and a direct pairwise subtraction of symmetric operating voltages returned $\Omega_{\oplus,\mathrm{sub}} = 7421(853)\,\mu\mathrm{rad\,s^{-1}}$. Both estimates overshoot the expected Bonn projection of $56.46\,\mu\mathrm{rad\,s^{-1}}$ by more than two orders of magnitude. We attribute this drastic discrepancy to direction-independent systematic offsets that currently dominate the measurement.

Furthermore, we estimated the lock-in threshold at $\Omega_L \lesssim 3.60(22) \times 10^{-3}\,\mathrm{rad\,s^{-1}}$, which places the Sagnac beat generated by Earth's rotation deep inside the unresolvable lock-in region. This is reflected in the stark deviations of the experimentally determined earth rotation rates mentioned above.

IX. ACKNOWLEDGMENTS

We acknowledge the usage of Gemini 3.1 [7] for drafting support, text refinement, and LaTeX formatting assistance during the preparation of this manuscript as well as coding support from Claude Code from Anthropic [8] and Codex [9] from OpenAI.

Appendix A: Raw time-series measured with an iPad

Figure 18 shows the complete unsmoothed one-hour gyroscope dataset used for the Allan-deviation and spectral analysis done in Section II B . The time series provides an overview of the raw sensor behavior over the full measurement interval and allows visual identification of offsets, slow variations, and possible disturbances before further statistical processing.

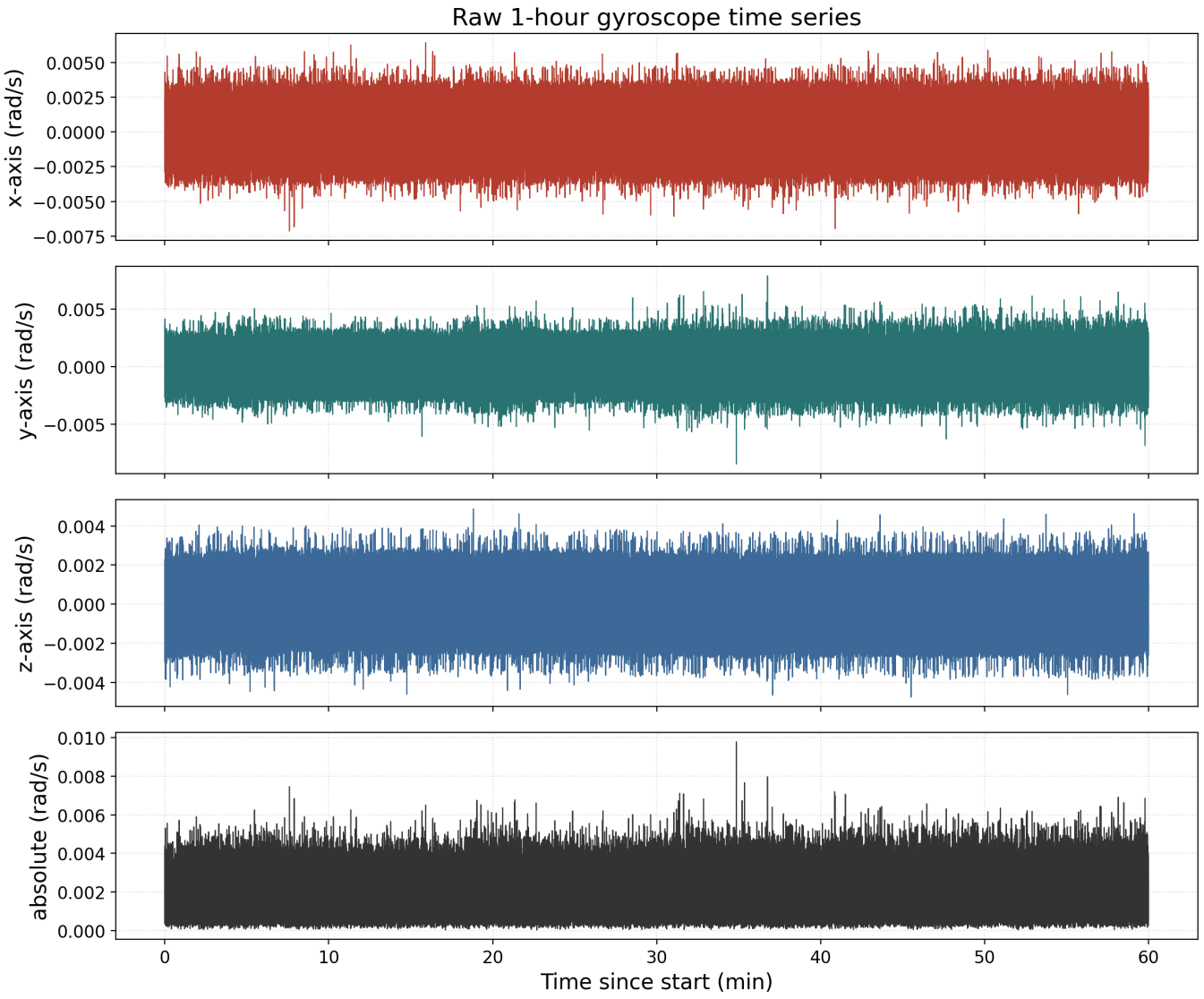


FIG. 18: Raw one-hour gyroscope time series of the iPad measurement. Shown are the unsmoothed angular-velocity components along the x-, y-, and z-axes, as well as the absolute angular rate $|\boldsymbol{\omega}| = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$, over the full measurement duration.

Appendix B: Lock-Monitor File Selection

Figures 19 and 20 map the file-by-file lock-monitor median for the positive and negative rotation directions, respectively. Each file is plotted against its sequential index within its motor-voltage folder. Green markers denote files retained by the threshold established in Equation (14), while red crosses indicate rejected files. The dashed line designates the empirical threshold at 0.0294 V. The deep-unlock branch clustering near 0.018 V is successfully isolated and rejected, whereas the functional branches near 0.042 V, 0.094 V and 0.123 V are safely retained. To maintain analytical symmetry, the same threshold derived from the positive branch is applied uniformly to the negative direction data (Figure 20).

It is worth noting that while error bars representing the statistical uncertainty of each file's median voltage are technically rendered on these plots, they remain visually obscured. The exceptionally high number of data samples acquired within each individual file shrinks the statistical variance to such a degree that the resulting standard error of the mean is microscopically small compared to the physical footprint of the data markers themselves.

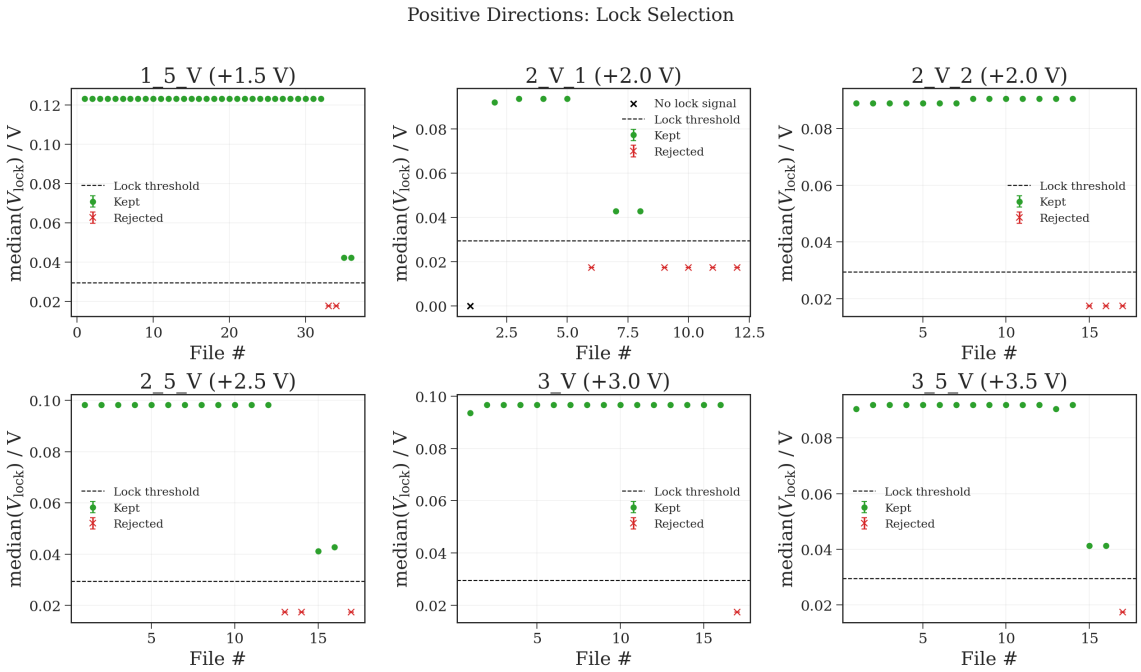


FIG. 19: File-by-file lock-monitor voltage for the positive rotation direction. Green circles represent files retained by Equation (14), while red crosses mark files in the deep-unlock branch near 0.018 V. The dashed line denotes the threshold at 0.0294 V. The statistical error bars on individual data points are technically plotted but remain completely invisible, as the extensive sample size per file suppresses the variance far below the graphical bounds of the markers.

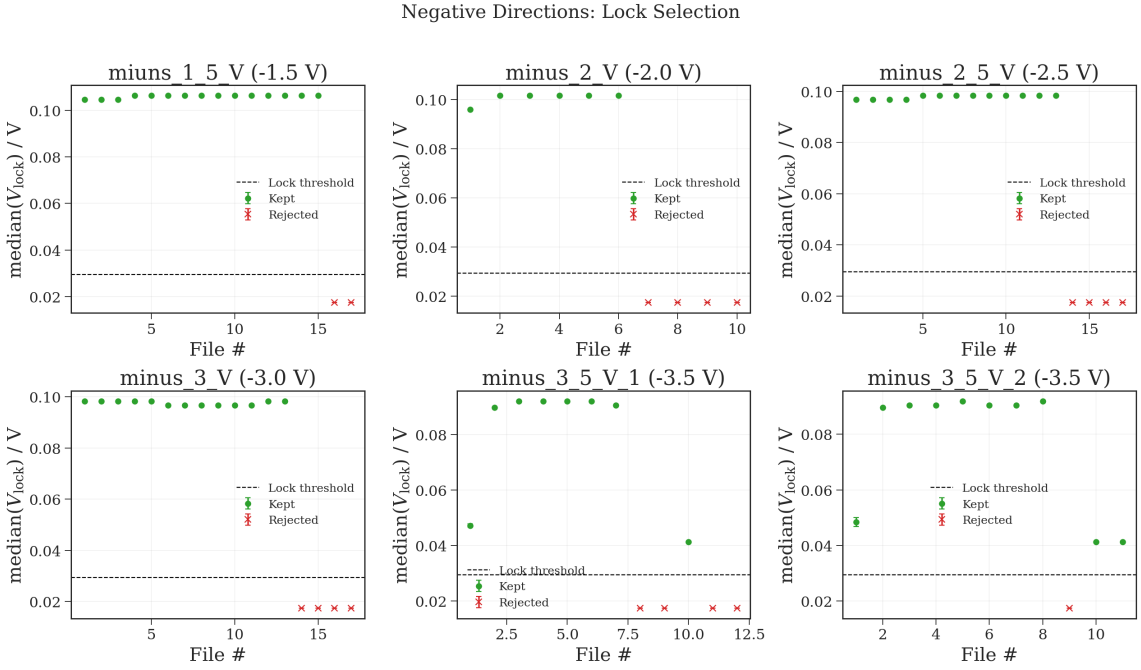


FIG. 20: File-by-file lock-monitor voltage for the negative rotation direction. The same threshold derived from the positive branch is applied here to ensure both directions are treated symmetrically. Similar to the positive direction, individual error bars are imperceptible due to the high statistical precision achieved by the large volume of intra-file measurements.

Appendix C: Per-Folder Sagnac Beat Distributions

Figures 21 and 22 present the file-wise mean Sagnac beat frequency $\langle \delta\nu \rangle$ across the accepted files for each specific motor-voltage setting. The green shaded band represents a one Standard Error of the Mean (SEM) confidence interval around the folder's global mean. Because no filtering cut was applied directly to the beat frequency itself, the retained data organically span the expected operational ranges without introducing artificial gaps that could bias the final Sagnac estimator. The folder means and their corresponding aggregate SEMs provide the exact y-axis coordinates utilized in Table III.

As with the lock-monitor data, while error bars tracking the individual per-file standard errors are plotted on the graphs, they are visually imperceptible. The massive volume of instantaneous frequency samples acquired during each measurement run drastically reduces the internal standard error, rendering the specific statistical uncertainties far smaller than the visual markers.

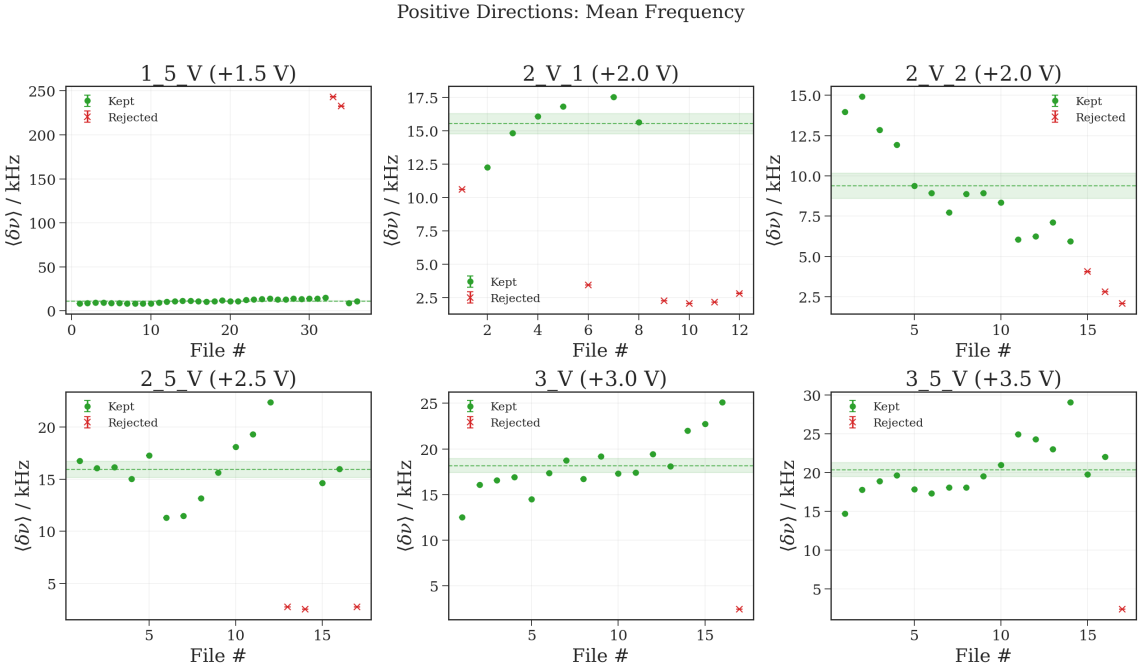


FIG. 21: Mean Sagnac beat frequency per accepted file, positive direction. The green band dictates ± 1 SEM around the grouped folder mean. These collective folder means supply the positive signed $\delta\nu$ entries found in Table III. The internal error bars on the individual data points are plotted but remain completely obscured by the markers due to the massive sample size heavily suppressing the standard error of the mean.

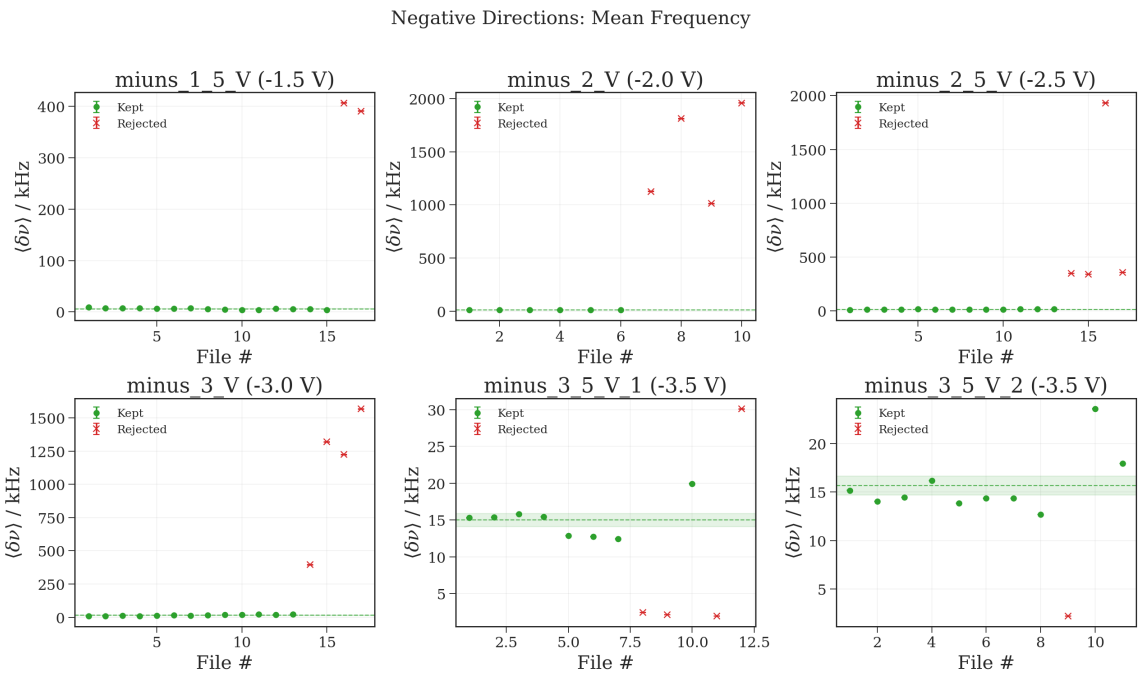


FIG. 22: Mean Sagnac beat frequency per accepted file, negative direction. Together with Figure 21, this establishes the full signed dataset basis for the fits evaluated in Sec. IV. Similar to the positive direction, individual file-level error bars are rendered invisible by the vast volume of samples minimizing the internal standard error.

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